

Tree-Level Calibration Is Not Plot-Level Calibration: Uncertainty Propagation Pitfalls in Drone-Based Agroforestry Carbon Estimation

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Abstract

Drone-based measurement, reporting and verification (MRV) is lowering the cost of bringing smallholder agroforestry into carbon markets. A typical pipeline predicts each tree’s diameter at breast height (DBH) from drone-measured crown size and height, maps DBH to biomass through a convex allometric equation, sums to a plot total, and applies a conservative deduction before issuing credits. We study the uncertainty-handling steps of this pipeline on an open ground-truth dataset of 6,292 trees from semi-arid agroforestry in western India, across seven DBH models ranging from a pooled Gaussian process to hierarchical Bayesian models with partial pooling over species. We report three findings. (i) Back-transforming a point estimate of $\log(\text{DBH})$ through the allometry gives plot totals 14–17% below per-sample propagation for every model; the classical $\exp((b\sigma)^2/2)$ correction predicts the gap to within 2 percentage points. (ii) Prediction intervals that are well calibrated per tree are badly over-confident per plot: nominal 80% intervals contain the true plot total for only 46–67% of pseudo-plots, because plot-level errors are systematic rather than averaging out. As a result, issuing credits at a posterior lower quantile (p_{25}) over-claims on 46–58% of plots, and a flat 80% haircut over-claims on 19–41% while forgoing 13–17% of credits in aggregate. (iii) Hierarchical modelling reduces per-tree error by 17% only when species is known; with drone-observable covariates alone it matches the pooled Gaussian process. We release code and a checklist for practitioners.

Keywords: carbon MRV; agroforestry; uncertainty quantification; calibration; hierarchical Bayesian models; Gaussian processes

1. Introduction

Tree planting on smallholder farms combines climate mitigation with income diversification for farmers in climate-vulnerable regions. Carbon finance could pay for it, but only if the carbon stored can be measured, reported and verified (MRV) at a cost proportional to small, fragmented plots. Field inventories do not meet that bar. Drone imagery combined with machine learning does: a detector finds trees in an orthomosaic, crown size and height are read from the imagery and a digital elevation model, a regression model predicts each tree’s diameter at breast height (DBH), and an allometric equation converts DBH into biomass and CO₂ (Gupta et al., 2025).

Most of the attention in such pipelines goes to the learned components: detection and DBH regression. This paper is about what happens *after* the DBH model — how its uncertainty is propagated to a plot total and turned into a credit claim. Those steps are a few lines of code each, but they decide how many credits are issued, and errors in them are invisible in the per-tree accuracy metrics that model development usually tracks.

We study three questions on an open agroforestry ground-truth dataset ([Farmers for Forests, 2025](#)), using seven DBH models with different covariates and inductive biases, so that each conclusion can be checked for model dependence:

1. **Back-transform.** How large is the bias from pushing a point estimate of $\log(\text{DBH})$ through a convex allometry, and is it model-dependent?
2. **Plot-level calibration.** Do intervals that are calibrated per tree remain calibrated for plot totals, which is the unit at which credits are issued? How do common issuance policies — a flat fractional haircut, or a posterior lower quantile — behave at the plot level?
3. **Covariates.** How much of the accuracy gain from species-aware hierarchical modelling survives when only drone-observable covariates are available?

Contributions. (i) A same-split comparison of seven DBH models with matched metrics, including a covariate ablation separating what is deployable from imagery alone (§3). (ii) Empirical confirmation across all seven models that the point-estimate back-transform under-reports plot CO_2 by 14–17%, matched by the closed-form lognormal correction (§4). The correction is classical in forest allometry ([Baskerville, 1972](#); [Sprugel, 1983](#)); our point is that it re-enters ML pipelines whenever the upstream model’s predictive distribution is collapsed to a point. (iii) A pseudo-plot evaluation showing that per-tree calibration does not transfer to plot totals, so that neither a flat haircut nor a raw posterior quantile is a safe issuance rule (§5). (iv) A short practitioner checklist (§6).

2. Setting, Data and Models

2.1. From DBH to credits

We use the allometric chain of the open drone-MRV methodology of [Gupta et al. \(2025\)](#): a multi-species power law for above-ground biomass (AGB, kg) taken from Verra-listed literature — Eq. 3.2.1 of [Brown \(1997\)](#), fitted on 28 trees from dry forest in India; total biomass adds roots (root-to-shoot ratio 0.27); carbon is 50% of dry mass; and CO_2 is 44/12 of carbon, following the Verra VMD0001 module ([Verra, 2023](#)):

$$\text{AGB} = e^a \text{DBH}^b, \quad a = -1.996, \quad b = 2.32, \quad \text{CO}_2 = 1.27 \times 0.5 \times \frac{44}{12} \times \text{AGB}. \quad (1)$$

All pipelines in this paper predict $\log(\text{DBH})$, as is standard for allometric quantities. A plot’s claimed CO_2 is then a functional of the predictive distribution of the plot total $T = \sum_{i \in \text{plot}} \text{CO}_{2i}$. We compare five issuance policies: the *point* total (each tree’s median DBH pushed through Eq. 1); the *mean* of T under per-sample propagation; the posterior lower quantiles $p_{25}(T)$ and $p_{10}(T)$; and a flat *80% rule*, $0.8 \mathbb{E}[T]$, the conservative policy recommended in an open drone-MRV methodology ([Gupta et al., 2025](#)).

2.2. Data

We use the public ground-truth release of 13 October 2025 ([Farmers for Forests, 2025](#)): 7,326 field-measured trees from four districts of Maharashtra, India, each with species, DBH,

height, two crown widths, plantation year, field-recorded age, block and GPS location. We drop implausible records ($\text{DBH} \leq 1$ or ≥ 200 cm, $\text{height} \leq 0.5$ m, $\text{crown} \leq 0.3$ m, $\text{DBH} < 2$ cm with $\text{height} > 3$ m) and records without age; exclude bamboo (72 records), to which the hardwood allometry does not apply; and keep species with at least 30 trees. The modelling set has **6,292 trees from 18 species** (median DBH 8.0 cm, range 1.3–53.5 cm). It is geographically concentrated: 92% of trees are in one district. We use an 80/20 split stratified by species (5,035 / 1,257 trees); every model in this paper is trained and scored on this same split.

Pseudo-plots. The release has no plot identifier. To evaluate plot-level behaviour we group test trees into pseudo-plots in two ways: *block-year* (district $\times$ block $\times$ plantation year; 26 groups with ≥ 10 test trees, covering 1,208 trees, median 25 trees per group) and *geo-cell* (GPS rounded to 10^{-3} degrees, ≈ 110 m; 39 groups, 894 trees, median 20). Block-year is our primary grouping; geo-cell checks robustness.

2.3. DBH models

Pooled Gaussian process (GPR). Following the specification of the open methodology above (Gupta et al., 2025), we fit a GP (Rasmussen and Williams, 2006) on standardised crown size and height with a scaled RBF kernel plus white noise, target $\log(\text{DBH})$, using scikit-learn (Pedregosa et al., 2011). A variant on log inputs, and a pooled log–log ordinary least squares (OLS) model, serve as further pooled baselines.

Hierarchical Bayesian family. Writing $\tilde{x}$ for a log-covariate centred on its training mean, the full model (*H-full*) is

$$\begin{aligned} \log(\text{DBH}_i) &\sim \mathcal{N}(\mu_i, \sigma_{s_i}^2), \\ \mu_i &= \alpha_{s_i} + \beta_{s_i} \tilde{c}_i + \zeta_{s_i} \tilde{c}_i^2 + \gamma_{s_i} \tilde{h}_i + \eta_{s_i} m_{s_i} \tilde{a}_i + \delta_{d_i}, \end{aligned} \tag{2}$$

with s_i the species, d_i the district, c, h, a crown, height and age. Species coefficients are partially pooled (Gelman and Hill, 2007) towards global means in non-centred form; the age slope is constrained non-negative ($\eta_s = \eta_g + |\xi_s|$) so that older trees are not predicted thinner; $m_s = 0$ masks age for the two species whose within-species age coefficient of variation is below 0.10. Priors are listed in Appendix A. Three ablations switch terms off without other changes:

- *H-drone*: species pooled ($\alpha, \beta, \zeta, \gamma$ shared), district effect kept, no age — deployable from imagery plus location;
- *H-species*: species-level coefficients, no district, no age;
- *H-species+district*: species and district, no age.

We sample with NUTS (Hoffman and Gelman, 2014) in PyMC (Abril-Pla et al., 2023): 4 chains $\times$ 1,000 draws after 1,000 tuning steps, target acceptance 0.95. All runs have $\hat{R} \leq 1.01$ on every hyperparameter and no divergent transitions. All predictive distributions are represented by 1,000 joint samples over the test set; for the GPR we sample from the full joint predictive covariance, so that plot totals carry correlated error in every model.

Table 1: Per-tree accuracy on the same 1,257 test trees. “Drone-only” marks models whose inputs are observable from imagery plus location. Calibration slope is the slope of empirical against nominal coverage at levels 0.5–0.95 (1 is ideal; < 1 means under-confident).

Model	Inputs	Drone-only	RMSE (cm)	MAE (cm)	Cal. slope
GPR	crown, height	yes	2.87	1.75	0.86
GPR (log inputs)	crown, height	yes	2.67	1.73	0.86
Pooled OLS	crown, height	yes	2.84	1.86	0.89
H-drone	+ district	yes	2.83	1.81	0.87
H-species	+ species	no	2.34	1.57	0.90
H-species+district	+ species, district	no	2.39	1.57	0.89
H-full	+ species, district, age	no	2.24	1.48	0.88

3. What Covariates Buy

Table 1 separates the gain from the hierarchical structure from the gain from the covariates it can use. With drone-observable inputs only, the hierarchical model (H-drone, 2.83 cm) is indistinguishable from the pooled GP (2.67–2.87 cm) and OLS; the district effect adds nothing. Knowing the species reduces RMSE to 2.34 cm (a 17% reduction over H-drone), and field-recorded age reduces it a further 6% to 2.24 cm. District adds nothing on top of species. The practical reading is that partial pooling over species pays off *if* species can be identified from imagery; until then, the simpler pooled model is as accurate. All models are mildly under-confident per tree (90% intervals contain $\approx 93\%$ of test trees).

Per-tree accuracy is not the quantity that credits are issued on. The species-aware models without age over-predict the test-set CO₂ total by 9.2% under per-sample propagation, against 2–3% for the pooled models and 4.5% for H-full; lower per-tree error does not imply a less biased total. We return to plot totals in §5.

4. The Back-Transform Gap

If $\log(\text{DBH}) \sim \mathcal{N}(\mu, \sigma^2)$, then by Eq. 1

$$\frac{\mathbb{E}[\text{AGB}]}{e^a (e^\mu)^b} = \exp\left(\frac{(b\sigma)^2}{2}\right) = \exp(2.69\sigma^2) \quad (b = 2.32). \quad (3)$$

A pipeline that collapses each tree’s predictive distribution to its median (or to $e^{\hat{\mu}}$) before applying the allometry therefore under-reports biomass by this factor, whatever the model. The correction is well known for allometric regressions fitted in log space (Baskerville, 1972; Sprugel, 1983; Sileshi, 2014); the same bias re-appears in ML pipelines where σ is the *predictive* uncertainty of an upstream model and the allometry is applied downstream as a fixed post-processing step. Analytical approximations that linearise the allometry around the mean (the delta method) drop the same term.

Figure 1 and Table 2 show the effect on the test set. For every model the point-estimate total is 14–17% below the per-sample mean (gap 16–21% of the point total), and Eq. 3 at the

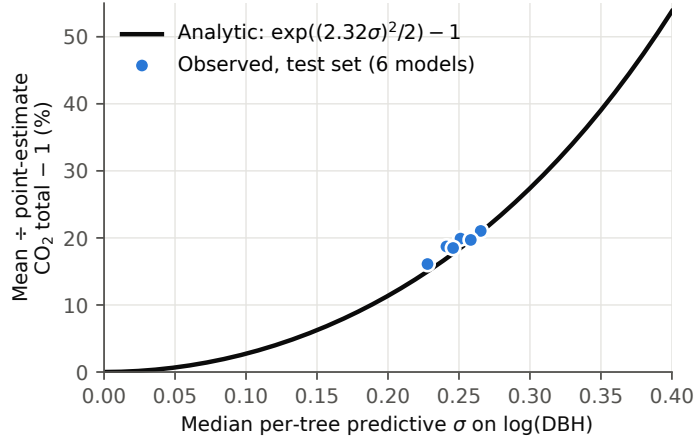


Figure 1: Gap between the per-sample mean and the point-estimate CO_2 total on the test set, for six models, against each model’s median per-tree predictive σ . The curve is Eq. 3.

Table 2: Test-set CO_2 total relative to field ground truth under the point-estimate and the per-sample (mean) back-transform, and the gap between them against the analytic prediction at the model’s median per-tree σ .

Model	Point	Mean	Gap (observed)	Median σ	Gap (Eq. 3)
GPR	−14.9%	+2.0%	19.9%	0.251	18.5%
Pooled OLS	−14.8%	+3.2%	21.0%	0.265	20.9%
H-drone	−12.1%	+5.2%	19.7%	0.258	19.7%
H-species	−8.0%	+9.2%	18.7%	0.241	16.9%
H-species+district	−7.8%	+9.2%	18.5%	0.246	17.7%
H-full	−10.0%	+4.5%	16.1%	0.228	15.0%

model’s median σ predicts the gap to within 2 percentage points. The point-estimate totals land 8–15% *below* ground truth. A pipeline that makes this choice is already conservative by roughly the size of a typical haircut, before any haircut is applied — and a practitioner comparing models by point-estimate totals would see the best per-tree model (H-full) as biased low by 10% when its per-sample total is within 5%. The fix requires no retraining: propagate samples, or apply Eq. 3 per tree.

5. Tree-Level Calibration Does Not Transfer to Plots

Credits are issued per plot, so the quantity that must be calibrated is the predictive distribution of the plot total T (Gneiting et al., 2007). Figure 2 compares coverage per tree and per plot for the pooled GPR and H-full. Per tree, both models are slightly under-confident. Per plot, both are strongly over-confident: nominal 80% intervals contain the true total for

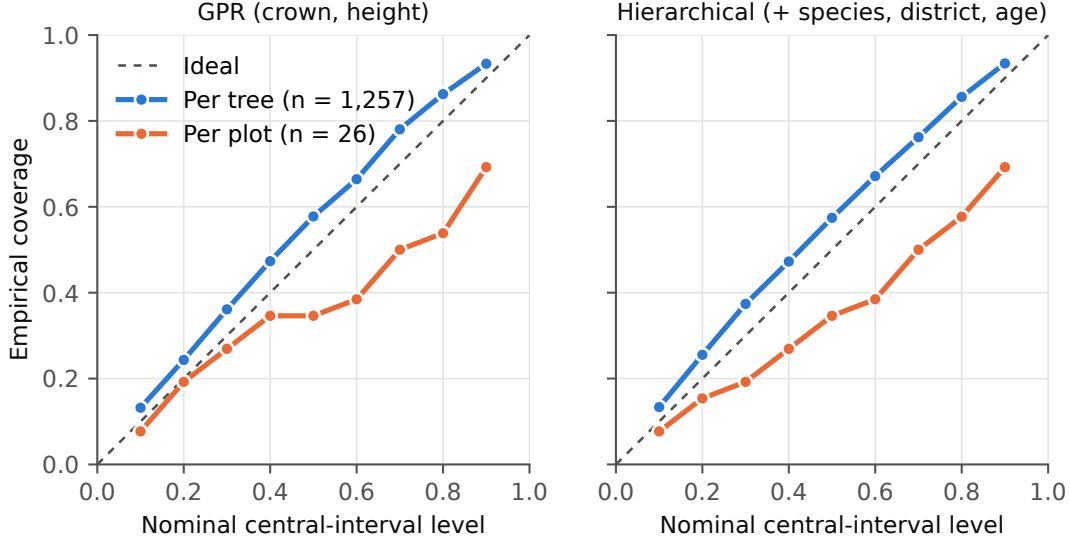


Figure 2: Empirical coverage of central predictive intervals for individual trees (blue) and for block-year pseudo-plot totals (orange). Both models are slightly under-confident per tree and strongly over-confident per plot.

54% (GPR) and 58% (H-full) of block-year plots. Across all five models and both groupings, 80% plot intervals cover 46–67% of plots and 50% intervals cover 19–44% (Appendix B).

The reason is visible in the per-plot errors. For H-full, the mean-policy error on block-year plots ranges from -35% to $+55\%$, and its magnitude is uncorrelated with plot size (Spearman $\rho = -0.07$). Errors that were independent between trees would shrink with plot size; these do not. They are shared within a plot — plausibly through management (orchard pruning), site conditions or measurement crews — and no model in this study has a term for them. Summing per-tree predictive distributions that are conditionally independent given the global parameters then yields a plot-total distribution that is far too narrow.

Consequences for issuance. Table 3 scores the policies plot by plot. A posterior lower quantile is an attractive replacement for a flat haircut, since it adapts the deduction to each plot’s uncertainty and has a clear interpretation — but only if the plot-level distribution is calibrated. Here it is not: p_{25} over-claims on 46–58% of plots instead of 25%, and p_{10} on 31–42% instead of 10%. The flat 80% rule is more conservative in aggregate — it forgoes 13–17% of true CO_2 — yet still over-claims on 19–35% of plots, because plot errors larger than 20% are common. The geo-cell grouping gives the same picture (p_{25} over-claims on 41–54%, the 80% rule on 20–41%; Appendix B). The hierarchical model with all covariates has the smallest plot-level error (mean absolute error 14.7% under p_{25} , against 19.2% for GPR), so covariates help, but they do not fix the calibration.

Neither policy is safe, for the same reason: both treat the model’s tree-level uncertainty as if it described the plot. The remedy is to calibrate at the level of issuance — for example a plot-level random effect fitted on surveyed plots, or a conformal or empirical quantile

Table 3: Issuance policies on 26 block-year pseudo-plots. Over-claim: share of plots where the claim exceeds the field-measured total (95% Wilson interval for H-full in brackets). MAE: mean absolute plot error of the policy. Left / over: total under- and over-shoot across plots as a share of true CO₂.

Model	Over-claim rate				80% rule
	mean	p_{25}	p_{10}	80% rule	left / over
GPR	65%	46%	31%	23%	16.7% / 1.4%
H-drone	62%	54%	42%	31%	15.7% / 4.0%
H-species	77%	58%	31%	19%	13.3% / 1.2%
H-species+district	77%	54%	42%	35%	14.3% / 3.6%
H-full	65%	46%	31%	19%	14.5% / 1.5%
	[46–81]	[29–65]	[17–50]	[9–38]	
Nominal target	—	25%	10%	—	

adjustment estimated on held-out plots ([Angelopoulos and Bates, 2023](#)) — and to check over-claim rates against the nominal level on plots the model has not seen.

6. Discussion

Checklist for practitioners.

1. *Propagate samples, not points.* Push predictive samples of $\log(\text{DBH})$ through the full allometric chain and sum per sample; or apply Eq. 3 per tree. Compare totals between models only after doing so.
2. *Validate at the level of issuance.* Report coverage and over-claim rates of plot totals on held-out plots, not per-tree coverage.
3. *Model shared plot error.* Include a plot- or farm-level effect, or calibrate the plot-total quantile empirically, before using it for credits.
4. *Split by plot.* Tree-level random splits place trees from the same farm in training and test, which flattens plot-level error.
5. *Separate deployable from field covariates.* Report accuracy with the inputs available at inference time.

Limitations. Our plots are pseudo-plots built from administrative and spatial groupings, and there are few of them (26 and 39), so rates carry wide intervals (Table 3). Because the split is by tree, test plots share farms with training trees; error on genuinely unseen plots is likely larger, which would strengthen the plot-level conclusion. The data come from one semi-arid region (92% from one district) and contain only two test trees above 50 cm DBH. Ground truth is itself the allometry applied to field DBH, so allometric error ([Chave et al., 2014](#)) and tree-detection error are outside our scope. The allometry is also used outside its stated domain: it was fitted for DBH 5–40 cm and is recommended for dry zones with more

than 900 mm/year of rainfall (Brown, 1997), whereas 92% of our trees come from a district with a normal annual rainfall of 561.6 mm (ICAR-CRIDA, 2011). This affects absolute CO₂ levels but not our conclusions. Re-running the full evaluation with the companion equation that Brown (1997) recommends below 900 mm ($AGB = 10^{-0.535} BA$, i.e. $b = 2$; Appendix C) shrinks the back-transform gap to 12–14%, as Eq. 3 predicts, while p_{25} still over-claims on 46–54% of block-year plots and nominal 80% plot intervals still cover only 46–65%. Saplings (<5 cm, 15% of test trees) carry about 1% of CO₂ but have large relative measurement noise.

Conclusion. The uncertainty-handling steps after the DBH model — back-transform, aggregation and issuance — matter as much as the model. On open agroforestry data, the choice of back-transform moves plot totals by 14–17% regardless of model, and intervals calibrated per tree under-cover plot totals by a wide margin, which makes both flat haircuts and naive posterior quantiles unreliable issuance rules. Both checks are cheap and should be part of any drone-based MRV pipeline.

Code and data. Code, saved posteriors and per-plot results: <https://anonymous.4open.science/r/XXXX> The ground-truth data are the public release of Farmers for Forests (2025), used without modification.

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Appendix A. Priors

Global means: $\alpha_g \sim \mathcal{N}(2.5, 0.5^2)$, $\beta_g \sim \mathcal{N}(0.5, 0.3^2)$, $\zeta_g \sim \mathcal{N}(0, 0.2^2)$, $\gamma_g \sim \mathcal{N}(0.3, 0.3^2)$, $\eta_g \sim \mathcal{N}^+(0.2, 0.3^2)$ (truncated at 0). Species scales: $\tau_\alpha \sim \mathcal{HN}(0.5)$, $\tau_\beta \sim \mathcal{HN}(0.3)$, $\tau_\zeta \sim \mathcal{HN}(0.15)$, $\tau_\gamma \sim \mathcal{HN}(0.3)$, $\tau_\eta \sim \mathcal{HN}(0.2)$. Species coefficients (non-centred): $\alpha_s = \alpha_g + \tau_\alpha z_{\alpha,s}$ and likewise for β, ζ, γ , with $z \sim \mathcal{N}(0, 1)$; age slope $\eta_s = \eta_g + \xi_s$, $\xi_s \sim \mathcal{HN}(\tau_\eta)$. District effects $\delta_d \sim \mathcal{N}(0, \tau_\delta^2)$, $\tau_\delta \sim \mathcal{HN}(0.3)$. Observation noise $\sigma_s \sim \mathcal{HN}(0.5)$ per species (a single σ when species is pooled). Ablations drop the corresponding terms and priors.

Appendix B. Robustness: Geo-Cell Pseudo-Plots and Coverage

Appendix C. Robustness: Low-Rainfall Allometry

Table 4: Over-claim rate by policy on 39 geo-cell pseudo-plots (≈ 110 m cells), and central-interval coverage of plot totals for both groupings (block-year / geo-cell).

Model	Over-claim rate (geo-cell)				Plot coverage	
	mean	p_{25}	p_{10}	80% rule	50%	80%
GPR	77%	46%	36%	28%	35 / 41%	54 / 59%
H-drone	72%	54%	44%	41%	19 / 26%	46 / 46%
H-species	72%	44%	26%	28%	23 / 44%	62 / 67%
H-species+district	67%	46%	26%	28%	27 / 38%	50 / 67%
H-full	67%	41%	28%	21%	35 / 38%	58 / 62%

Table 5: The evaluation of §4–5 repeated with Eq. 3.2.2 of Brown (1997) ($\text{AGB} = 10^{-0.535} \text{BA}$, BA in cm^2 ; $b = 2$), recommended for dry zones below 900 mm/year, against the main-text allometry (Eq. 3.2.1, $b = 2.32$). Block-year pseudo-plots. Gap: per-sample mean over point-estimate total, minus one.

Model	Gap		Over-claim rate, $b = 2$			80% coverage
	$b = 2.32$	$b = 2$	p_{25}	p_{10}	80% rule	$b = 2.32$ / $b = 2$
GPR	19.3%	13.9%	46%	31%	19%	54 / 58%
H-drone	19.8%	14.3%	54%	42%	27%	46 / 46%
H-species	18.6%	13.3%	46%	27%	15%	62 / 65%
H-species+district	18.5%	13.2%	54%	38%	27%	50 / 54%
H-full	16.0%	11.5%	46%	23%	15%	58 / 65%
Nominal target	—	—	25%	10%	—	80%